Project report part B TET4205 Group N

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I. Introduction

This report presents Part B of the course project in TET4205 Power System Analysis 2. It expands on the DCPF solver implemented in Part A, with the motivation of mimicking a real life Optimal Power Flow (OPF) analysis. Operating a grid, transmission system operators are required to follow the N-1 criterion [1, slide 12]. This security is often associated with a cost. This part of the project aims to find this cost. DC-OPF and preventive DC-SCOPF were implemented, and analyzed using a contingency analysis to determine the cost of security.

II. Methodology and Implementation

The methodology for Part B of the project builds on the *Direct Current Power Flow* (DCPF) implemented in Part A. The assumptions $R \ll X \rightarrow G \approx 0$, $|V| \approx 1$ and $\sin(\delta_i - \delta_j) \approx \delta_i - \delta_j$, $\cos(\delta_i - \delta_j) \approx 1$ still apply [1, slide 20]. First, *DC Optimal Power Flow* (DC-OPF) is applied to the chosen grid. Second, a contingency analysis is performed to identify line outages that violate the N-1 criterion. Third, *DC Security-Constrained Optimal Power Flow* (DC-SCOPF) is applied to find a cost-optimal, N-1 secure operating point. Finally, another contingency analysis is applied to verify the results [2, slide 6]. This was implemented in Python, using the framework from Part A. DC-OPF and DC-SCOPF were formulated in Pyomo, and solved using HiGHS [1, slide 15], [2, slide 10]. The contingency was built using the DCPF solver from Part A.

A. DC Optimal Power Flow

DC-OPF uses linear programming to minimize the total generation cost. DC-OPF doesn't account for any security constraints needed to fulfill the N-1 criterion, meaning that the resulting operating point is not guaranteed to be N-1 secure.

$$\min \sum_g c_g P_{G,g} \tag{1}$$

$$\mathbf{B}\boldsymbol{\delta} = \mathbf{P}_G - \mathbf{P}_D, \quad \delta_{slack} = 0 \tag{2}$$

$$P_{G,g}^{min} \leq P_{G,g} \leq P_{G,g}^{max}, \quad -P_{ij}^{max} \leq \frac{\delta_i - \delta_j}{X_{ij}} \leq P_{ij}^{max} \tag{3}$$

Where c_g is the marginal cost for generator g, $P_{G,g}$ is the active power generation from generator g, $P_{D,i}$ is the demand at bus i, δ_i is the voltage angle at bus i, X_{ij} is the reactance between buses i and j, $\mathbf{B}$ is the susceptance matrix from Part A and P_{ij}^{max} is the thermal limit of the line between i and j [1, slide 30].

The objective function from Equation (1) minimizes the total generation cost. The decision variables are the generator outputs $\mathbf{P}_G$ and the bus voltage angles $\boldsymbol{\delta}$, while the demand $\mathbf{P}_D$ is given. The power balance shown in Equation (2) describes the DCPF from Part A, ensuring that the net injection at each bus is equal to the power flowing out [1, slides 20–21]. The slack bus is fixed as a reference. The constraints from Equation (3) keep each generator within its operating range, and each line flow below its thermal limit. Since both the objective function and the constraints are linear, the problem is formulated as a linear programming problem. Any optimal solution is therefore a global optimum [1, slides 26, 28].

B. Contingency Analysis

The contingency analysis has the purpose of checking whether the DC-OPF point of operation fulfills the N-1 criterion [1, slide 13]. This is performed on the values calculated from the DC-OPF meaning $\mathbf{P}_G$ is the same in all contingency cases. Every line is removed one by one, and the $\mathbf{B}$ is built for every new iteration of the grid. The new matrix is called $\mathbf{B^{(k)}}$ and is used to solve for new values for all remaining lines with DCPF. A line is considered overloaded if $|P_{ij}^{(k)}| > P_{ij}^{\max}$, where the absolute value accounts for changing power flow direction. Outages causing one or more overloads are defined as critical, and collected in $\mathcal{K}$, which is the basis for DC-SCOPF. Only physical lines are evaluated, not the artificial generator connections [2, slide 15].

C. Preventive DC-SCOPF

Preventive DC-SCOPF minimizes generation cost while enforcing operating limits in both the intact network and the critical contingency cases identified in the initial contingency analysis. A single generation vector $\mathbf{P}_G$ is shared across cases, while each case k has its own voltage-angle vector $\boldsymbol{\delta}^{(k)}$. The optimization therefore includes [2, slides 23–25]:

$$\begin{aligned} &\mathbf{B}^{(k)}\boldsymbol{\delta}^{(k)} = \mathbf{P}_G - \mathbf{P}_D, \\ &\left|P_{ij}^{(k)}\right| \leq P_{ij}^{\max}, \end{aligned} \qquad \forall k \in \{0\} \cup \mathcal{K}. \tag{4}$$

where $\mathcal{K}$ is the set of critical line outages identified in the initial contingency analysis, consisting of lines 1165 and 1288. The capacity constraints apply to all remaining transmission lines and transformers in each contingency case. Generator limits and a reference angle are also enforced. The resulting dispatch requires no corrective redispatch following an outage.

III. Results, Scenario and Assumptions

The network topology from Part A is retained, while generation capacities, marginal costs and branch limits are adjusted to construct the Part B scenario. The changes mainly involved replacing the specified generators, which were originally based on actual power generation units, with representative generators that represent different categories of power generation. These categories include hydropower, wind and nuclear power.

Both optimisation models use identical input data. Generator minimum outputs are assumed to be zero, and branch ratings are treated as active-power limits under the DC approximation. Only single physical transmission-line outages are considered.

A. DC-OPF Results

The DC-OPF solution supplies the total demand of 2720 MW at a generation cost of 33,110 €/h. Wind and nuclear generators operate at their maximum capacities, while hydro generation covers the remaining demand. Consequently, Vann_161 is the marginal generating unit, resulting in a uniform locational marginal price (LMP) of 31.5 €/MWh across all buses. No transmission constraints are binding in the base case, with the highest branch loading being 76.6% on line 1174. Thus, the system operates within all transmission limits under normal conditions.

B. Contingency Analysis of the OPF Solution

The contingency analysis evaluates the N-1 security of the DC-OPF operating point while keeping generator dispatch fixed. Out of 22 investigated line outages, two result in transmission overloads. The outage of line 1288 causes line 1174 to reach 104.5% loading, while the outage of line 1165 overloads line 6798 to 101.0%. The remaining contingencies do not result in overloads. This shows that although the DC-OPF solution satisfies all constraints under normal operating conditions, it does not meet the N-1 security criterion. This motivates the implementation of preventive DC-SCOPF.

C. Preventive DC-SCOPF

Preventive DC-SCOPF increases operating cost to approximately 33 449 €/h while maintaining total generation at 2720 MW. The output of `Vann_161` decreases from 620.1 to 533.5 MW, while `Vann_141` and `Vann_106` produce 50.0 and 36.6 MW. Other generator outputs remain unchanged. This redistribution increases cost by approximately 339 €/h or 1.02%. The generator dispatch is provided in Table IV in the appendix.

TABLE I
Comparison of DC-OPF and preventive DC-SCOPF.

Quantity	DC-OPF	DC-SCOPF
Generation [MW]	2720	2720
Operating cost [€/h]	33 110	33 449
Maximum N0 loading [%]	76.6	72.4
Maximum N-1 loading [%]	104.5	100.0
Outages causing overload	2	0

D. Verification of N-1 Security

To verify the DC-SCOPF solution, a new contingency analysis was performed using the optimised generator dispatch. The generator outputs were kept constant, while a separate DCPF calculation was performed for the intact network and each of the 22 line outages. Both transmission lines and transformers were checked against their respective capacity limits. The results are presented in the appendix in Table VII. It shows that no component exceeds its capacity limit in any of the investigated cases. In the intact network (N0), line 1174 has the highest loading at 72.4 %. The two most critical contingencies are the outages of lines 1165 and 1288. When line 1165 is disconnected, transformer 141–109 reaches 100% loading. Similarly, the outage of line 1288 causes line 1174 to reach 100% loading. These results indicate that the capacity constraints are active in both cases. The verification therefore confirms that the preventive DC-SCOPF solution satisfies the N-1 criterion for all single line outages considered.

IV. Discussion

The results demonstrate that an economically optimal operating point under normal conditions does not necessarily satisfy N-1 security [2, slide 13]. The DC-OPF dispatch follows the merit order, with wind and nuclear generation operating at maximum capacity and hydro acting as the marginal generation source.

Preventive DC-SCOPF addresses the identified overloads by redistributing 86.6 MW of hydro generation from bus 161 to buses 141 and 106. This reduces loading in critical contingency scenarios, at an additional generation cost of approximately 1.02%. The two most critical contingencies reach exactly 100% loading after SCOPF, meaning that the solution satisfies the thermal limits but has no additional security margin in these cases.

The results depend on the assumed generation costs, capacities and branch limits, which were adjusted to create a meaningful test scenario. Furthermore, the preventive formulation keeps generation dispatch fixed across all contingencies, requiring security to be maintained without post-outage redispatch [2, slides 14, 32]. Finally, the DC approximation neglects power losses, reactive power and voltage magnitude variations [2, slide 4]. An AC assessment would therefore be required to evaluate these aspects of system security.

V. Conclusion

DC-OPF supplied the 2720 MW demand at approximately 33 110 €/h without violating intact-network limits. However, two of the 22 investigated line outages caused overloads. Preventive DC-SCOPF redistributed generation to address the two critical contingencies. Subsequent verification confirmed that the resulting dispatch satisfied the branch-flow limits for all 22 investigated line outages. Operating cost increased to approximately 33 449 €/h, giving a cost of security of 339 €/h, or 1.02%. Thus, preventive DC-SCOPF achieved N-1 secure operation within the considered DC model at a modest additional cost.

VI. Appendix

TABLE II
OPTIMAL GENERATOR DISPATCH FOR THE DC-OPF BASE CASE.

Generator	Bus	Cost [€/MWh]	P_g [MW]	$P_g^{\max}$ [MW]
Vind_45	45	1.0	112.5	112.5
Nuclear_45	45	10.5	189.5	189.5
Vind_108	108	1.5	150.0	150.0
Nuclear_108	108	9.8	252.6	252.6
Vann_161	161	31.5	620.1	677.0
Vind_161	161	0.8	187.5	187.5
Nuclear_161	161	10.2	252.6	252.6
Vind_62	62	2.0	60.0	60.0
Nuclear_62	62	11.5	94.7	94.7
Vind_176	176	1.2	225.0	225.0
Nuclear_176	176	10.0	189.5	189.5
Vind_30	30	1.8	75.0	75.0
Nuclear_30	30	11.0	126.3	126.3
Vind_80	80	1.4	90.0	90.0
Nuclear_80	80	10.8	94.7	94.7
Total production cost				33,110 €/h
Total generation				2,720 MW

TABLE III
BRANCH FLOWS FROM THE DC-OPF SOLUTION, SORTED BY LOADING.

Branch	**Flow (MW)**	**Loading (%)**
Line 1174 (106–76)	−574.3	76.6
Line 6798 (45–30)	−470.3	62.7
Line 1288 (138–176)	−421.1	56.2
Transformer 161–177	416.3	55.5
Line 1175 (60–45)	−409.2	54.6
Transformer 141–109	−406.1	54.1
Line 6562 (61–79)	486.1	54.0
Line 1292 (76–60)	−379.3	50.6
Transformer 45–61	363.1	48.4
Line 1176 (109–79)	−426.1	47.3
Line 1317 (78–108)	−422.5	46.9
Line 6799 (177–30)	416.3	46.3
Line 1289 (138–105)	341.1	45.5
Line 1165 (141–161)	−393.9	43.8
Transformer 76–78	−299.5	39.9
Transformer 46–179	196.6	26.2
Line 6801 (179–108)	196.6	21.8
Line 1336 (107–80)	−170.1	18.9
Line 1166 (77–107)	−170.1	18.9
Transformer 46–178	−117.3	15.6
Transformer 176–180	−106.6	14.2
Line 1293 (78–61)	123.0	13.7
Line 6800 (30–178)	117.3	13.0
Line 6802 (108–180)	106.6	11.8
Line 1290 (105–139)	75.7	10.1
Line 1334 (46–62)	−79.3	8.8
Line 1173 (105–76)	−64.5	8.6
Transformer 77–60	−29.9	4.0
Line 1291 (139–106)	−24.3	3.2
Line 1335 (62–80)	25.4	2.8

TABLE IV
GENERATOR DISPATCH FOR PREVENTIVE DC-SCOPF.

Generator	Bus	Cost [€/MWh]	P_G [MW]	$P_G^{\max}$ [MW]
Vind_45	45	1.0	112.5	112.5
Nuclear_45	45	10.5	189.5	189.5
Vind_108	108	1.5	150.0	150.0
Nuclear_108	108	9.8	252.6	252.6
Vann_161	161	31.5	533.5	677.0
Vind_161	161	0.8	187.5	187.5
Nuclear_161	161	10.2	252.6	252.6
Vind_62	62	2.0	60.0	60.0
Nuclear_62	62	11.5	94.7	94.7
Vind_176	176	1.2	225.0	225.0
Nuclear_176	176	10.0	189.5	189.5
Vind_30	30	1.8	75.0	75.0
Nuclear_30	30	11.0	126.3	126.3
Vind_80	80	1.4	90.0	90.0
Nuclear_80	80	10.8	94.7	94.7
Vann_141	141	35.0	50.0	500.0
Vann_106	106	36.0	36.6	250.0

TABLE V
MOST CRITICAL N-1 CONTINGENCIES WITH FIXED GENERATOR DISPATCH.

Outage	Most loaded line	Loading [%]	Flow [MW]
1288 (138–176)	1174 (106–76)	104.5	-783.6
1165 (141–161)	6798 (45–30)	101.0	-757.6
1289 (138–105)	1174 (106–76)	99.2	-743.9
6562 (61–79)	1165 (141–161)	97.8	-880.0
1176 (109–79)	1165 (141–161)	91.1	-820.0
6799 (177–30)	1165 (141–161)	90.0	-810.2

TABLE VI
MOST CRITICAL N-1 CONTINGENCIES AFTER PREVENTIVE DC-SCOPF.

Line outage	Most loaded branch	Loading [%]	Flow [MW]
1165 (141–161)	Transformer 141–109	100.0	-750.0
1288 (138–176)	Line 1174 (106–76)	100.0	-750.0
1289 (138–105)	Line 1174 (106–76)	94.7	-710.2
6562 (61–79)	Line 1165 (141–161)	92.2	-830.0
1176 (109–79)	Line 1165 (141–161)	85.6	-770.0
1317 (78–108)	Line 1288 (138–176)	84.7	-635.6

TABLE VII
N-1 VERIFICATION RESULTS FOR THE DC-SCOPF SOLUTION.

Contingency	Most loaded component	Loading (%)
N0	Line 1174	72.40
Line 1166	Line 1174	70.45
Line 1334	Line 1174	73.37
Line 1335	Line 1174	72.15
Line 1336	Line 1174	70.45
Line 1165	Transformer 141–109	**100.00**
Line 1289	Line 1174	94.70
Line 1290	Line 1174	81.79
Line 1291	Line 1174	68.46
Line 1173	Line 1174	79.29
Line 1174	Line 1290	81.79
Line 1292	Transformer 45–61	74.95
Line 1175	Transformer 45–61	80.29
Line 6798	Line 1174	69.09
Line 1288	Line 1174	**100.00**
Line 6562	Line 1165	92.22
Line 1293	Line 1174	71.44
Line 1176	Line 1165	85.56
Line 1317	Line 1288	84.74
Line 6799	Line 1165	80.40
Line 6800	Line 1174	73.28
Line 6801	Line 1174	74.96
Line 6802	Line 1174	79.16

VII. INDIVIDUAL CONTRIBUTIONS

The individual contributions to the project are summarized below. The contributions are based on the work performed by each group member and the corresponding development history.

- **Andreas Pedersen Monsen:** Responsible for programming and implementation of DC OPF and report writing.
- **Sverre Prøsch-Moen:** Responsible for implementation of DC-SCOPF and project writing.
- **Tor Kristian Boiesen:** Responsible for the Excel spreadsheet and its interpretation.
- **Håkon Nordal Grenne:** Responsible for implementation of DC OPF and contingency analysis.
- **Theo Bråten:** Responsible for writing project report and editing.
- **Vegard Ernstsen Gresslien:** Responsible for verification of DC-SCOPF and report writing.

VIII. USE OF ARTIFICIAL INTELLIGENCE

Generative AI tools were used as supplementary support during the project. This included assistance with understanding DC-OPF and DC-SCOPF concepts, discussing the mathematical formulation, interpreting optimization results, and reviewing and improving parts of the Python implementation. AI was also used to improve the clarity and language of the report. All modelling decisions, code implementation, testing and interpretation of results were reviewed and evaluated by the group members, who remain responsible for the submitted work and its conclusions.

REFERENCES

[1] D. Baltensperger, "Prerequisites for security-constrained optimal power flow (SCOPF)," lecture slides, TET4205 Power System Analysis 2, NTNU, Trondheim, Norway, Oct. 2026.

[2] D. Baltensperger, "Security-constrained optimal power flow (DC formulation)," lecture slides, TET4205 Power System Analysis 2, NTNU, Trondheim, Norway, Oct. 20, 2026.